Prathamesh Patil

Backend Engineer · Node.js · TypeScript · Python · SQL

+91 73490 41840 | prathameshpatil330@gmail.com | prathameshpatil.vercel.app | github.com/Prathameshh-Patil | LinkedIn

SUMMARY

Backend engineer (B.E. AI & ML, Oct 2026) with 11 months of internship experience building REST APIs in Node.js/TypeScript and FastAPI on MySQL and PostgreSQL, shipped with Docker, CI/CD and AWS. I focus on correctness under load: transactions, indexes and tests that fail when a rule breaks. Based in Bengaluru, available to join immediately.

EXPERIENCE

AI & Cloud Computing Intern, SuprMentr Technology Pvt Ltd — Nov 2025 – Mar 2026

- Contributed to a multi-agent RAG platform built with LangGraph, including a self-evaluation step that tracks hallucination rate and answer accuracy.
- Wrote PostgreSQL analytics queries (window functions, CTEs) powering weekly usage dashboards in React.

Software Engineer Intern, AI Division, Jmedia Corp — Jan 2025 – Jun 2025

- Built FastAPI REST APIs serving LLM-backed product features, secured with JWT authentication and rate limiting.
- Containerised the services with Docker, automated build and deployment with GitHub Actions CI/CD, and monitored latency and error rates on Grafana dashboards.

PROJECTS

Portfolio Platform: REST API, Website & Admin Panel — API docs | GitHub

TypeScript, Node.js, Express, MySQL 8.4, Drizzle ORM, Zod, Next.js, Docker, AWS Lambda + RDS, GitHub Actions

- Built an Express + MySQL REST API (routes, services, repositories) with shared Zod contracts, so the Next.js frontend and API fail to compile if they disagree.
- Generated OpenAPI 3.1 docs from the same Zod schemas the routes validate with; an integration test checks real responses against the spec.
- Designed composite MySQL indexes verified with EXPLAIN: on 200,000 rows, paginated inbox queries fell from 26.6–35 ms to 0.23–0.26 ms (100–150× faster).
- Deployed on AWS Lambda + API Gateway with RDS MySQL (TLS, Secrets Manager) from one SAM template; CI runs 33 API tests on a real MySQL container and 13 Playwright E2E tests per push.

Slotwise: Event Booking Platform — Live | GitHub

FastAPI, PostgreSQL, SQLAlchemy, Celery, Redis, Next.js, TypeScript, Docker

- Prevented double booking with a partial unique index and single-transaction orders; a 12-thread race test asserts exactly one request wins a seat.
- Implemented 10-minute seat holds and all-or-nothing multi-seat orders; moved hold expiry and confirmation emails to Celery + Redis jobs with retries.

ReplyDesk: AI Support Inbox with Human Approval — Live | GitHub

FastAPI, PostgreSQL, Pydantic, Next.js, TypeScript, LLM API (Groq), Playwright

- Built a support inbox where four LLM agents sort, extract, draft and check each reply and a person approves it before sending; outputs are schema-validated and failures go to a manual queue.
- Raised category accuracy from 90% to 97% using an eval over 60 labelled messages; 84 API tests run against PostgreSQL.

SKILLS

Backend: Node.js, Express, FastAPI, REST API design, OpenAPI/Swagger, JWT auth, Zod, Pydantic, Celery
Databases: MySQL, PostgreSQL, Redis, Drizzle ORM, SQLAlchemy, indexing & EXPLAIN, transactions & locking
Cloud & DevOps: AWS (Lambda, API Gateway, RDS, SAM), Docker, GitHub Actions CI/CD, Grafana, Git, Linux
Languages: TypeScript, JavaScript, Python, SQL
Testing & Frontend: Vitest, Supertest, pytest, Playwright; React, Next.js, Tailwind CSS

EDUCATION

B.E., Artificial Intelligence & Machine Learning — 2022 – Oct 2026

Dayananda Sagar Academy of Technology and Management (DSATM), Bengaluru · CGPA 7.6